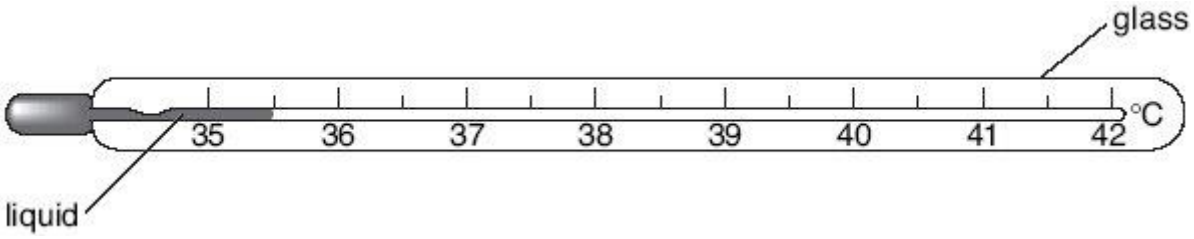
This practice paper covers the same content and skills as your actual assessment. It is a useful tool for evaluating how well your revision has gone.

**How to use:**

- Only try this paper after you have revised in other ways first
- Print the paper off
- Complete it under test conditions: in 45 minutes and with no other materials for help
- Mark yourself using the mark scheme at the end of this document
- Identify any weak areas and revise these parts again
- Try those questions again and see if you can improve your score

If you need help printing this resource or marking advice, feel free to ask your teacher!

**Q1.** The thermometer drawn below can be used to measure the temperature of the human body.




(a) (i) What is the lowest temperature **this** thermometer can measure?

.....°C

1 mark

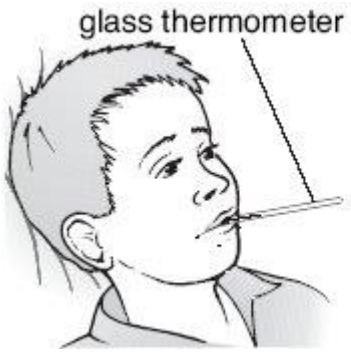
(ii) What is the normal temperature of the human body?
Tick the correct box.

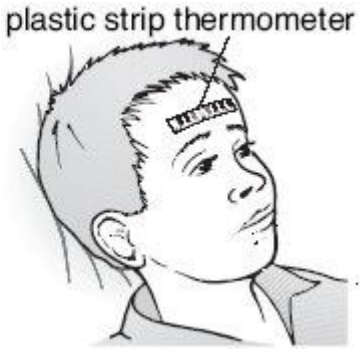
37°C ☐ 39°C ☐ 41°C ☐

1 mark

(iii) When we are ill our temperature may go up.

A nurse can measure a child's temperature with two different thermometers as shown below.



Give **one** reason why it is safer to use a plastic strip thermometer than a glass thermometer.

.....

.....

1 mark

(b) Viruses are micro-organisms that can make us ill.

Give the name of **one** other type of micro-organism that can make us ill.

.....

1 mark

(c) Alcohol and mercury are two liquids that can be used in glass thermometers. The table gives information about these liquids.

| liquid | boiling point (ºC) | colour |
|---|---|---|
| alcohol | 78 | colourless |
| mercury | 357 | shiny grey |

(i) A red dye is added to the colourless alcohol used in thermometers. Suggest a reason for this.

.....

.....

1 mark

(ii) Choose words from the list below to fill the gaps in the sentences.

**gas** **liquid** **solid**

When alcohol and mercury boil they both change from a liquid to a .....

1 mark

A thermometer containing mercury can be used to measure the temperature of an oven at 150°C because mercury is a ..... at 150°C.

1 mark

maximum 7 marks

**Q2.** Nancy is a dancer.



(a) When Nancy dances her arms and legs are moved by pairs of antagonistic muscles.

How do antagonistic muscle pairs work?
Tick the correct box.

Both muscles contract at the same time. ☐

One muscle is big and the other is small. ☐

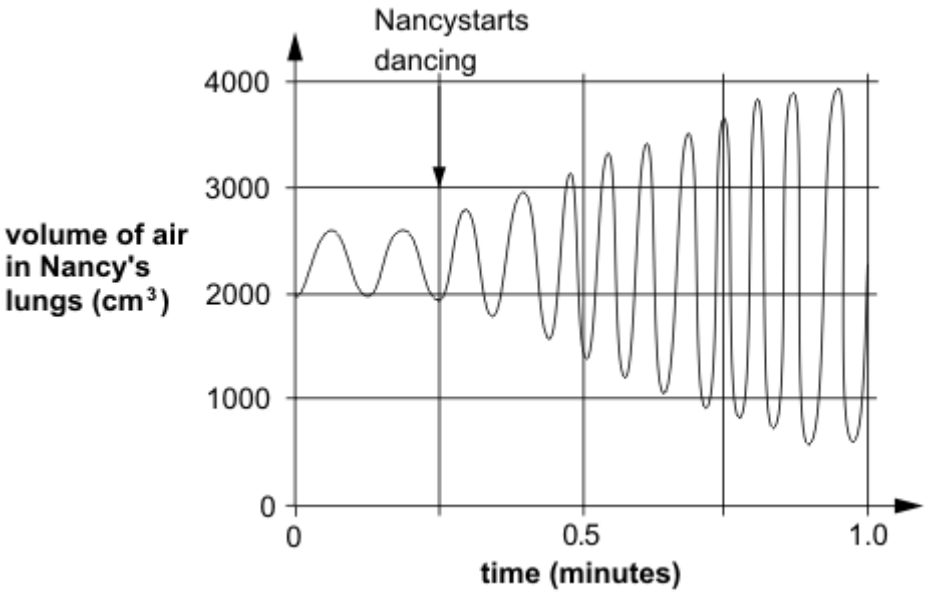
As one muscle contracts, the other relaxes. ☐

One muscle is strong and the other is weak. ☐

Both muscles relax at the same time. ☐

1 mark

(b) As Nancy dances her breathing changes because she needs more oxygen.
The graph below shows how the volume of air in her lungs changes when she dances.




From the graph, give **two** ways her breathing changes when she dances.

1. .....

1 mark

2. .....

1 mark

(c) Nancy's muscle cells produce carbon dioxide as she dances.

Which of the following shows how the carbon dioxide is removed from Nancy's body?
Tick the correct box.

muscle cells → bloodstream → windpipe → lungs → nose ☐

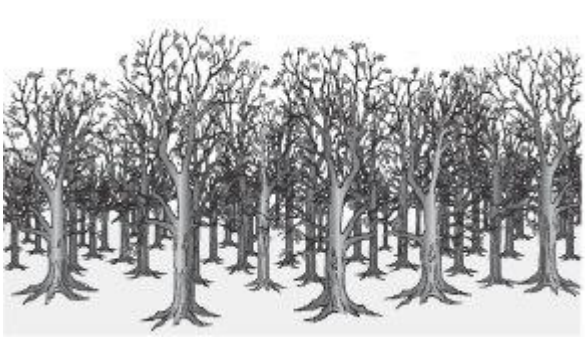
muscle cells → windpipe → lungs → bloodstream → nose ☐

muscle cells → bloodstream → lungs → windpipe → nose ☐

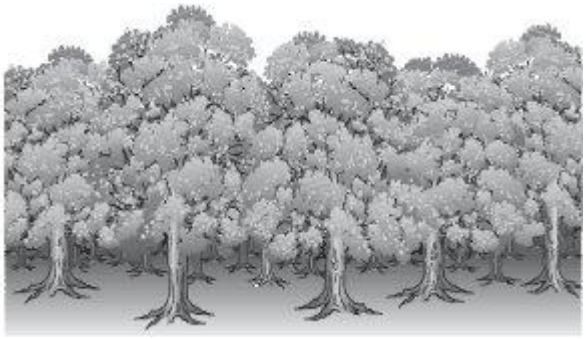
muscle cells → windpipe → bloodstream → lungs → nose ☐

1 mark
maximum 4 marks

**Q3.** The drawings below show the trees in a woodland area at the beginning of May and at the end of May.

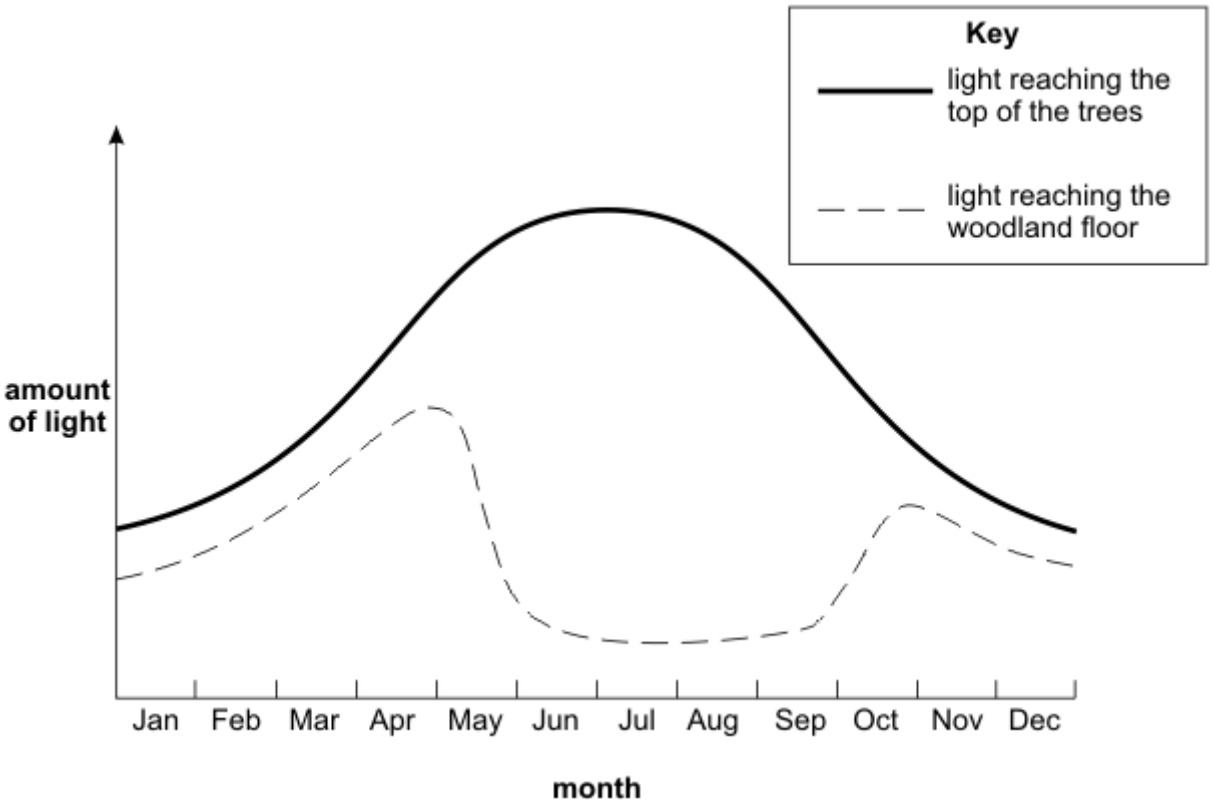


**beginning of May**



**end of May**

The graph below shows the amount of light reaching the top of the trees and the woodland floor over one year.




(a) Why does the amount of light reaching the woodland floor decrease during May?

.....

.....

1 mark

(b) Plants grow on the woodland floor.

Explain why these plants grow bigger **and** faster when there is plenty of light.

.....

.....

.....

.....

2 marks

(c) **Respiration** takes place in the cells of all plants.

Complete the word equation for **respiration**.

oxygen + ..... ⟶ carbon dioxide + .....

2 marks
maximum 5 marks

**Q4.** This question is about **three** different fuels, A, B and C.

Fuel A is stored in tanks. It is not stored under pressure. It flows along a pipe to where it is needed.

Fuel B is stored under pressure in small cylinders. It is used by campers.

Fuel C can be stored in sacks or bags.

(a) (i) Tick the correct box.

Fuel A is a:

solid ☐

liquid ☐

gas ☐

1 mark

Name a fuel which A could be .....

1 mark

(ii) Tick the correct box.

When fuel B comes out of the cylinder this is a:

solid ☐

liquid ☐

gas ☐

1 mark

Name a fuel which B could be .....

1 mark

(iii) Tick the correct box.

Fuel C is a:

solid ☐

liquid ☐

gas ☐

1 mark

Name a fuel which C could be .....

1 mark

(b) Complete the statement to describe what happens when a fuel burns.

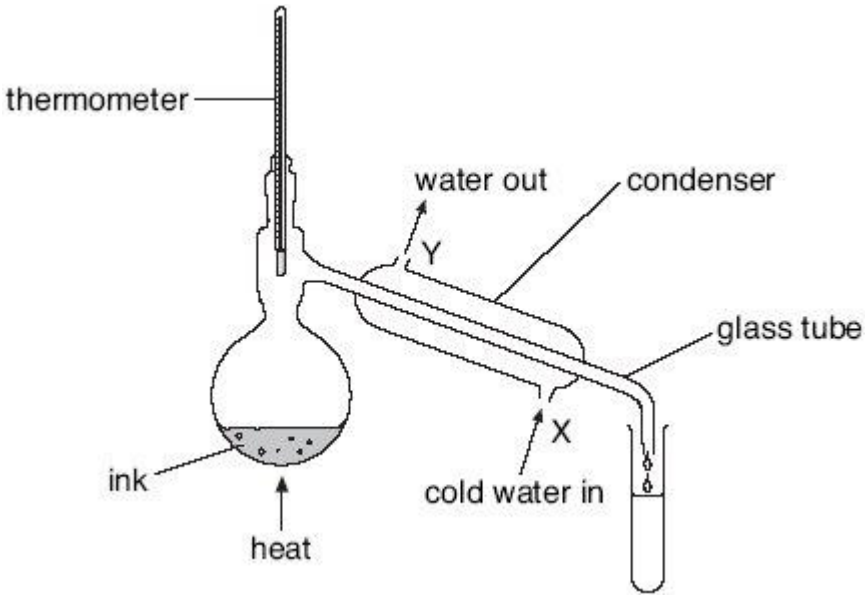
Two waste products formed from burning fuels are

..... and .....

2 marks

Maximum 8 marks

**Q5.** Rema used the apparatus below to distil 100 $cm^3$ of water-soluble ink.




**apparatus A**

*not to scale*

(a) Which processes occur during distillation?
Tick the correct box.

condensation then evaporation ☐

evaporation then condensation ☐

melting then boiling ☐

melting then evaporation ☐

1 mark

(b) Give the name of the colourless liquid that collects in the test-tube.

.....

1 mark

(c) What would the temperature reading be on the thermometer when the ink has been boiling for two minutes?

.....°C

1 mark

(d) (i) Water at 15°C enters the condenser at X.
Predict the temperature of the water when it leaves the condenser at Y.

.....°C

Explain this change of temperature.

.....

.....

1 mark

(ii) Give **two** ways in which the water vapour changes as it passes down the glass tube in the condenser.

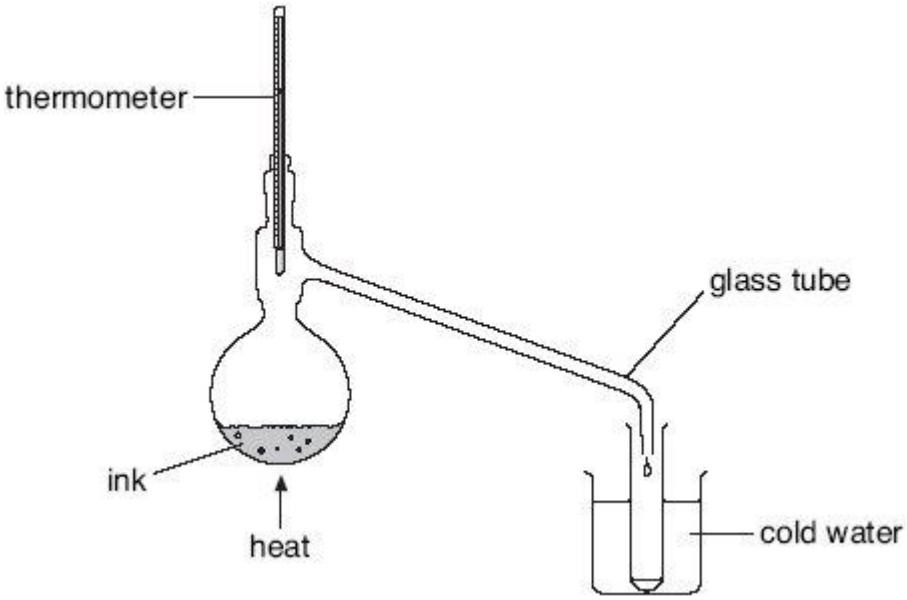
1. .....

1 mark

2. .....

1 mark

(e) Peter used the apparatus below to distil 100 $cm^3$ of water-soluble ink.




**apparatus B**

*not to scale*

Why is the condenser in **apparatus A** better than the glass tube and beaker of water in **apparatus B**?

.....

.....

1 mark
maximum 7 marks

**Q6.** (a) Many substances burn. Some of them are used as fuels. Some fuels are burned in power stations to generate electricity.

Tick the boxes by the **two** fuels which are **most often** burned in power stations.

coal ☐

paraffin wax ☐

natural gas ☐

petrol ☐

paper ☐

2 marks

(b) Burning fuels can pollute the air.
Give **two** substances which pollute the environment when fuels burn.

1 .....

2 .....

2 marks

(c) Some power stations do not burn fuels. They use other energy resources to generate electricity.

Give **two** energy resources which are **not** fuels and which are used to generate electricity.

1 .....

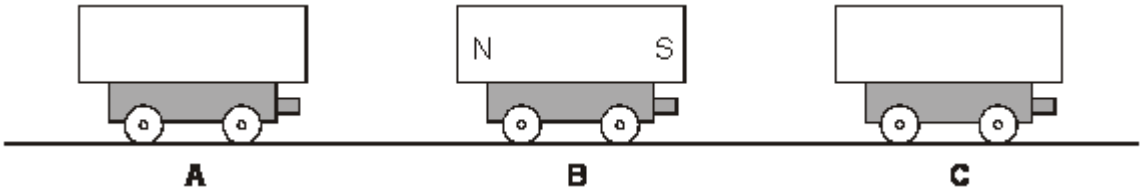
2 .....

2 marks
Maximum 6 marks

**Q7.** The diagram below shows three trolleys.
Peter put a bar magnet on each trolley.

(a) He pushed trolleys A, B and C together.

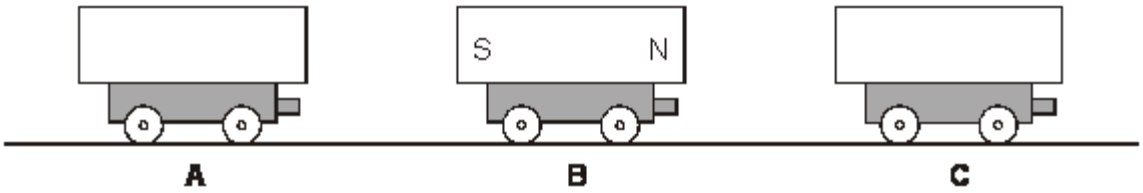
- Magnet B **attracted** magnet A.
- Magnet B **repelled** magnet C.




**On the diagram above**, label the north and south poles of magnets A and C.
Use the letters N and S.

2 marks

(b) Peter turned trolley B around. Trolleys A and C were **not** turned around.



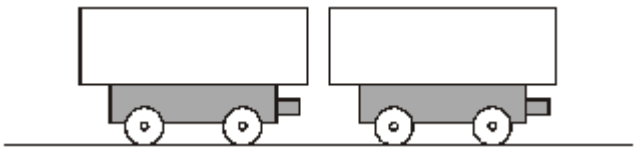

What would happen now when Peter pushed them all together?
Use either **attract** or **repel** to complete each sentence below.

Magnet B would ..... magnet A.

Magnet B would ..... magnet C.

1 mark

(c) Peter held two trolleys close together and then let go.



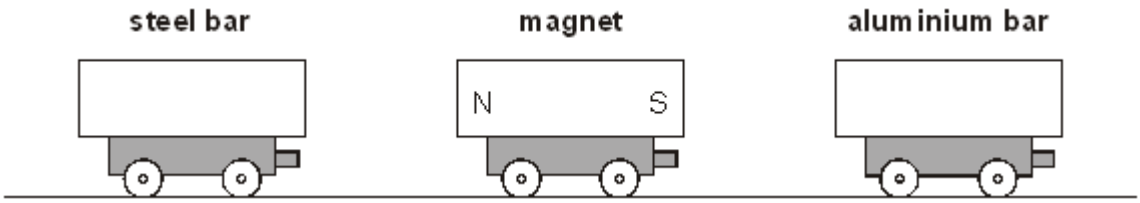
The magnets repelled each other.

**Draw an arrow** on both magnets to show which way they would move.

1 mark

(d) Peter took a magnet, a steel bar and an aluminium bar.

He put them on three trolleys as shown below.




(i) What happens to the steel bar as he moves it closer to the magnet?

.....

1 mark

(ii) What happens to the aluminium bar as he moves it closer to the magnet?

.....

1 mark

maximum 6 marks

Mark schemes

## Q1.

(a) (i) 35

**1 (L3)**

(ii) 37°C ✓

**1 (L4)**

(iii) any **one** from

- less chance of micro-organisms being passed on **or** spread
- glass could break
  *accept 'less chance of germs being passed on'*
  *accept 'might pass on disease'*
  *accept 'a child might bite **or** swallow the thermometer'*
  *if more than one box is ticked, award no mark*
- mercury **or** alcohol **or** liquid could spill (if the glass thermometer broke)
- mercury is poisonous
- you could choke

**1 (L3)**

(b) any **one** from

- bacteria
- fungi
- protozoa
  *accept a named non-viral pathogenic micro-organism such as 'salmonella'*
  *do **not** accept names of diseases*
  *do **not** accept 'germs'*

**1 (L4)**

(c) (i) any **one** from

- so that alcohol can be seen
  *accept 'so you can see it'*
- alcohol is difficult to see
- you cannot easily see a colourless liquid
  *accept 'you cannot see a colourless liquid'*
- to make it easier to read the thermometer
  *accept 'so you can read it'*

**1 (L4)**

(ii) • gas

**1 (L3)**

- liquid

**1 (L4)**

*answers must be in the correct order*

**[7]**

## Q2.

(a) As **one** muscle contracts, the other relaxes. ✓

*if more than one box is ticked, award no mark*

**1 (L5)**

(b) any **two** from

- she gets more air or oxygen

  *accept 'greater volume of air taken in **or** out'*
  *'she needs more oxygen' is insufficient*

  *accept 'breathes out more carbon dioxide'*
  *'shorter **or** heavier **or** louder **or** harder breaths' are insufficient*

- it gets faster

  *accept 'speeds up' or 'she takes more breaths'*
  *'it increases' is insufficient*

- it gets deeper

  *accept 'she takes bigger breaths'*
  *'the volume of air in her lungs increases' is insufficient*
  *do **not** accept 'the volume of her lungs increases'*

  *accept, for one mark, 'she breathes more' if not given with 'it gets faster' or 'it gets deeper'*

**2 (L5)**

(c)

| | |
|---|---|
| muscle cells → bloodstream → lungs → | ☑ |
| windpipe → nose | ☐ |

*if more than **one** box is ticked, award no mark*

**1 (L6)**

**[4]**

## Q3.

(a) • leaves are bigger

*accept 'there are leaves **or** a canopy'*
*accept 'leaves open **or** grow'*
*'trees block the light' is insufficient*

**1 (L5)**

(b) any **two** from

- light is needed for photosynthesis
  *accept 'photosynthesis'*
- more (photosynthesis)
  *accept, for two marks, 'the rate*
  ***or** amount of photosynthesis increases if the light increases'*
- light provides energy for growth
  *'they need light to grow' is insufficient*
  *'light is a source of food' is insufficient*
- biomass **or** food **or** sugar **or** starch **or** carbohydrate is produced

**2 (L6)**

(c) • glucose

**1 (L6)**

- water
  *answers must be in the correct order*

**1 (L6)**

**[5]**

## Q4.

(a) (i) liquid

*if more than one box is ticked award no mark*
*accept 'A' in the liquid box instead of a tick*

**1**

petrol **or** oil **or** diesel

*accept a brand name*
***or** paraffin **or** kerosene*

**1**

(ii) gas

*if more than one box is ticked award no mark*
*accept 'B' in the gas box instead of a tick*

**1**

gas **or** natural gas **or** calor

*accept a brand name*
*gas **or** butane **or** propane*
*do **not** accept 'hydrogen' **or** methane **or** camping gaz*

**1**

solid

*if more than one box is ticked award no mark*
*accept 'C' in the solid box instead of a tick*

**1**

coal **or** coke **or** wood **or** charcoal

*accept a brand name **or** 'peat'*

**1**

(b) any **two** from

- carbon dioxide
- water *accept 'steam'*
- smoke *accept 'soot'*
- ash
- gases *accept 'fumes'*
- sulphur dioxide *accept 'nitrogen oxides'*
- carbon monoxide

*do **not** accept 'heat' **or** 'energy' **or** 'lead'*

**2**

**[8]**

## Q5.

(a) evaporation then condensation ✓

*if more than one box is ticked, award no mark*

**1 (L5)**

(b) water

*accept 'distillate'*

**1 (L5)**

(c) 100

*accept a temperature from 99 to 101*

**1 (L5)**

(d) (i) a temperature above 15 but below 100

any **one** from

- it is heated by the water vapour
  *accept 'vapour **or** steam **or** hot water' for water vapour*
- thermal energy **or** heat is transferred from the water vapour
  *accept 'it heats up'*
  ***both** the answer and the correct explanation are required for the mark*

**1 (L6)**

(ii) any **one** from

- it condenses
  *accept 'it makes condensation'*
- it changes to a liquid
  *accept 'it is liquid'*
  *accept 'it changes state'*
  *accept 'it turns to water'*

**1 (L6)**

- its temperature falls
  *accept 'it cools' 'it changes temperature' is insufficient*

**1 (L6)**

(e) any **one** from

- cold water replaces warm water
  *accept 'it cools the water vapour better **or** more efficiently **or** quickly'*
  *accept 'it will work faster'*
- the water stays cold **or** cooler
- there is a bigger difference in temperature between the water vapour and the water
- some vapour escapes in B
  *accept 'it is cooled over a bigger length **or** for longer'*
  *accept 'more water is collected'*

**1 (L6)**

**[7]**

## Q6.

(a) coal ✓

**1 (L3)**

natural gas ✓

*if more than two boxes are ticked, award no mark*
*deduct one mark for each incorrectly ticked box*
*minimum mark zero*

**1 (L3)**

(b) any **two** from

- soot **or** carbon
- ash
- smoke
- sulphur dioxide
  *accept 'sulphur'*
- carbon monoxide
- oxides of nitrogen
  *accept 'nitrogen oxide'*
  *accept 'carbon dioxide*
  *accept 'lead'*
  *accept 'acid rain'*
  *do **not** accept 'greenhouse gases'* **or** *'water'* **or** *'gas'*

**2 (L3)**

(c) any **two** from

- geothermal

- wind
- solar

  *accept 'Sun'*
- running water

  *accept 'hydro'* ***or*** *'HEP'* ***or*** *'water power'*
- tidal

  *do* ***not*** *accept 'water'*
- waves

  *accept 'nuclear'* **or** *'uranium'*

**2 (L3)**

**[6]**

## Q7.

**Marks may be awarded for part (a) if the magnets are correctly labelled in part (b) and no answer is given in part (a)**

(a) • Magnet A




***both*** *poles are required for the mark*

**1 (L4)**

- Magnet C




***both*** *poles are required for the mark*

**1 (L4)**

(b) • repel

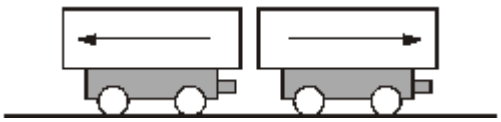
- attract

  *answers must be in the correct order*

  ***both*** *answers are required for the mark*

**1 (L4)**

(c) •



***both*** *arrows are required for the mark*

**1 (L4)**

(d) • it is attracted

*accept 'it gets faster'*

**1 (L4)**

- nothing

  *accept 'it is not attracted or repelled'*

  *accept 'it is not attracted'*

*accept 'it is not repelled'*
*'they stick together' is insufficient*
*do **not** accept 'it repels'*

**1 (L4)**

**[6]**